Detecting Pneumonia from Chest X-Rays Using Transfer Learning-Based CNNs

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Motivation & Research Question

Motivation

- Radiologists face immense workloads, leading to severe diagnostic delays and fatigue-induced errors.
- Automated initial X-ray screening serves as a rapid triage system—prioritizing critical findings, reducing physician burnout, and ensuring timely patient interventions.

Research & Hypothesis

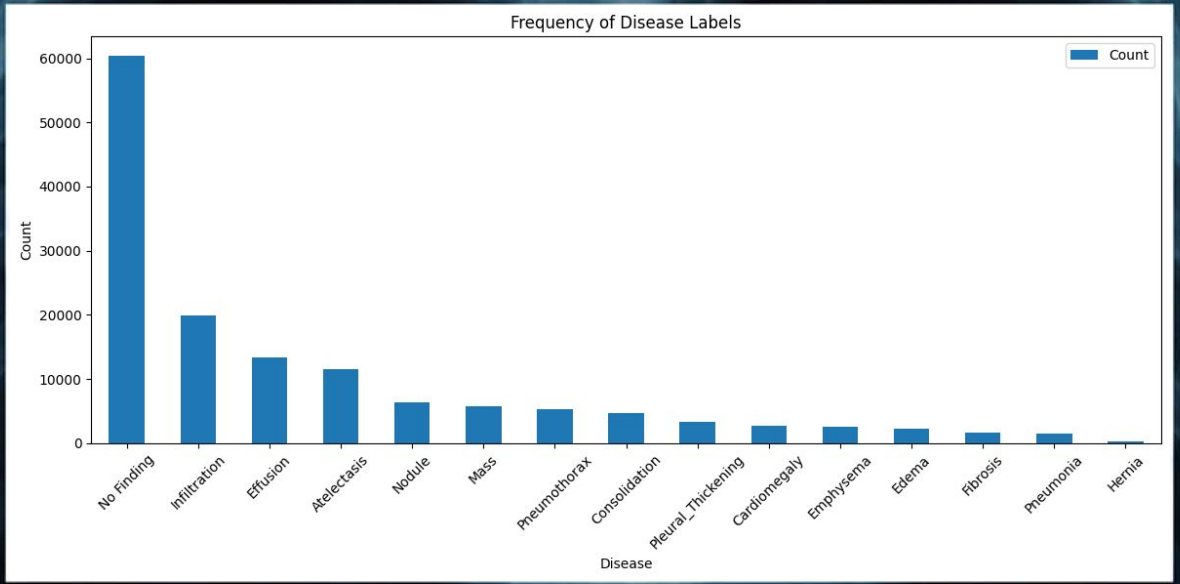
- Hypothesis: A transfer learning-based convolutional neural network, fine-tuned on the NIH ChestXray14 dataset, will accurately classify chest X-ray images for the presence or absence of disease, achieving strong performance across ROC-AUC, precision, recall, and F1 score metrics.
- Research Question: To what extent can a transfer learning-based convolutional neural network accurately detect disease from chest X-ray images, and how well does it perform perform across evaluation metrics such as ROC-AUC, precision, recall, and F1 score?

Data Acquisition & Explanation

We utilize the publicly accessible NIH ChestXray14 dataset via the NIH Clinical Center.

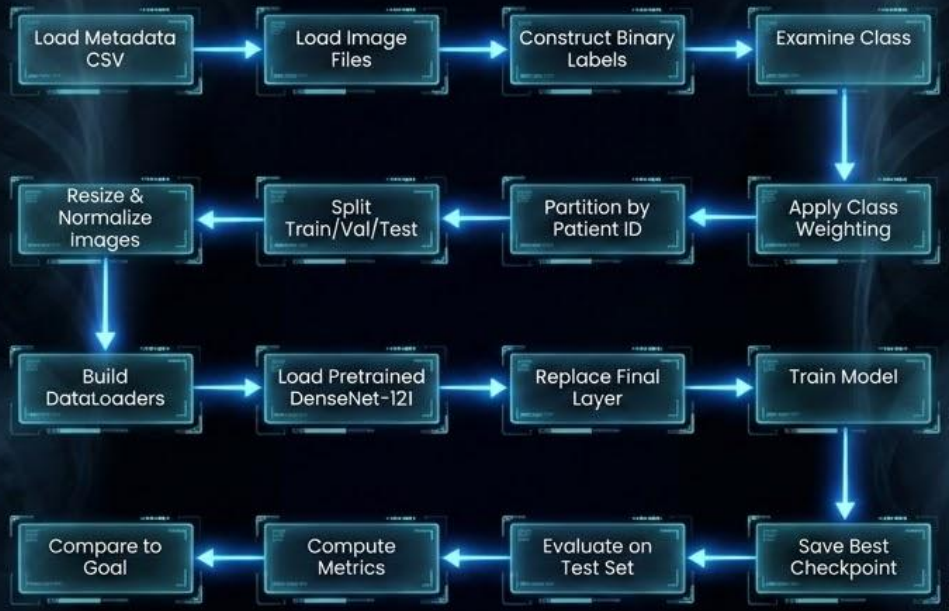
- Contains over 100,000 frontal chest radiographs.
- Labeled with findings across 14 distinct disease categories.
- Metadata pairs images with age, gender, view position, and pathology.
- Images will be preprocessed to fit standard sizing (224x224) and normalization.

Exploratory Analysis



Exploratory Insight: The data reveals a massive disparity in label distribution. Pneumonia, which was the original focus, accounts for less than 2% of the data. Training a model specifically for pneumonia would suffer from extreme majority-class bias, necessitating a strategic shift in our target variable.

Analysis Pipeline



Analysis Decision Making

Initial Approach

We initially focused strictly on detecting pneumonia. However, in the exploratory analysis it was revealed that it represents a very small minority of the labels.

The Problem

Severe class imbalance meant a model lazily predicting “No Finding” for every image would achieve misleadingly high accuracy without learning true diagnostic features.

The Pivot

We broadened the target to the presence of any “Finding” across all 14 categories. This balanced the data, allowing for meaningful ROC-AUC and F1 scores.

Bias & Uncertainty

Patient Leakage

Patients often have multiple scans. Random splitting could lead the model to memorize anatomy rather than disease markers.

Solution: Strict dataset partitioning by Patient ID.

Class Imbalance

Even with binary grouping, “No Finding” remains the majority class.

Solution: Apply class weighting within the BCEWithLogitsLoss function to heavily penalize minority-class errors.

Cohort Specificity

The model is trained entirely on NIH Clinical Center data.

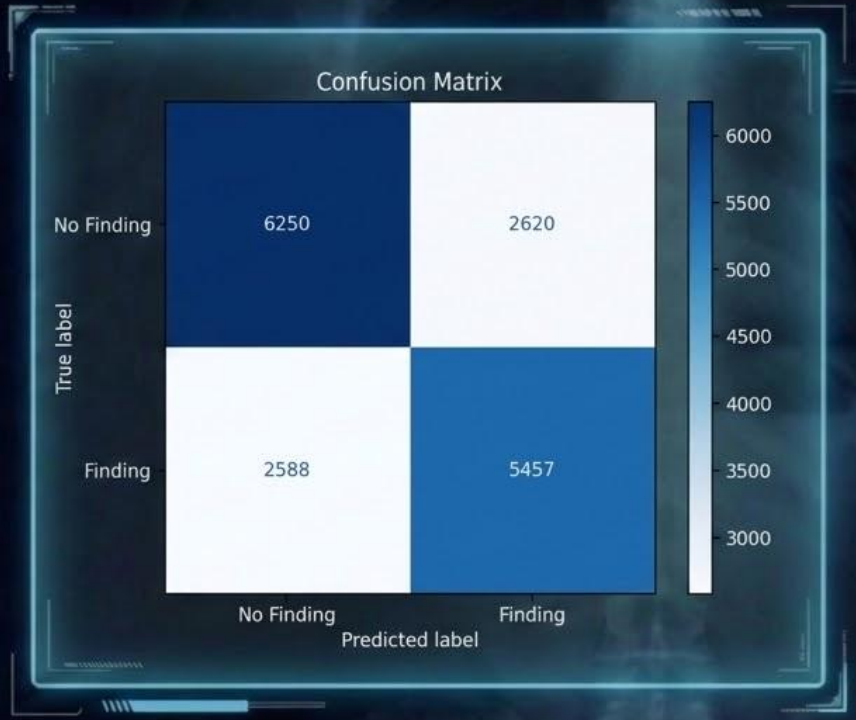
Limitation: Generalization performance may drop across hospitals with different scanning hardware or patient demographics.

Results: Confusion Matrix Analysis

Error Analysis

The model demonstrated balanced performance across both classes:

- True Negatives (6,250): High accuracy in identifying healthy lungs.
- False Negatives (2,588): Missed cases that require clinician backup.
- True Positives (5,457): Successfully flagged majority of pathologies.
- False Positives (2,620): Over-diagnoses that would lead to manual triage.



Final Results & Conclusions

0.7446
TEST ROC-AUC

0.6770
F1 SCORE

0.6756
PRECISION

0.6783
RECALL

Clinical Triage Utility: An AUROC of 0.74 demonstrates the model's high potential as a triage tool, reliably flagging 67.8% of pathologies while maintaining a solid specificity.

Data Validity: Our patient-level split ensures these results represent the model's ability to identify disease in new patients, not just recognize familiar anatomy.

The Verdict: DenseNet-121 is a robust architecture for binary classification in medical imraging. The balanced error rates suggest the model is well-calibrated for initial screening.

Next Steps

Model Interpretability: Integrate Grad-CAM (Gradient-weighted Class Activation Mapping) to produce visual heatmaps. This process identifies the specific radiographic regions that drive the model's classification, providing the transparency necessary to foster clinical confidence.

Multi-Label Expansion: Transition the framework from a binary 'Finding vs. No Finding' task to a complex multi-label system. Future iterations will focus on training the architecture to differentiate between and simultaneously detect the fourteen discrete thoracic pathologies.

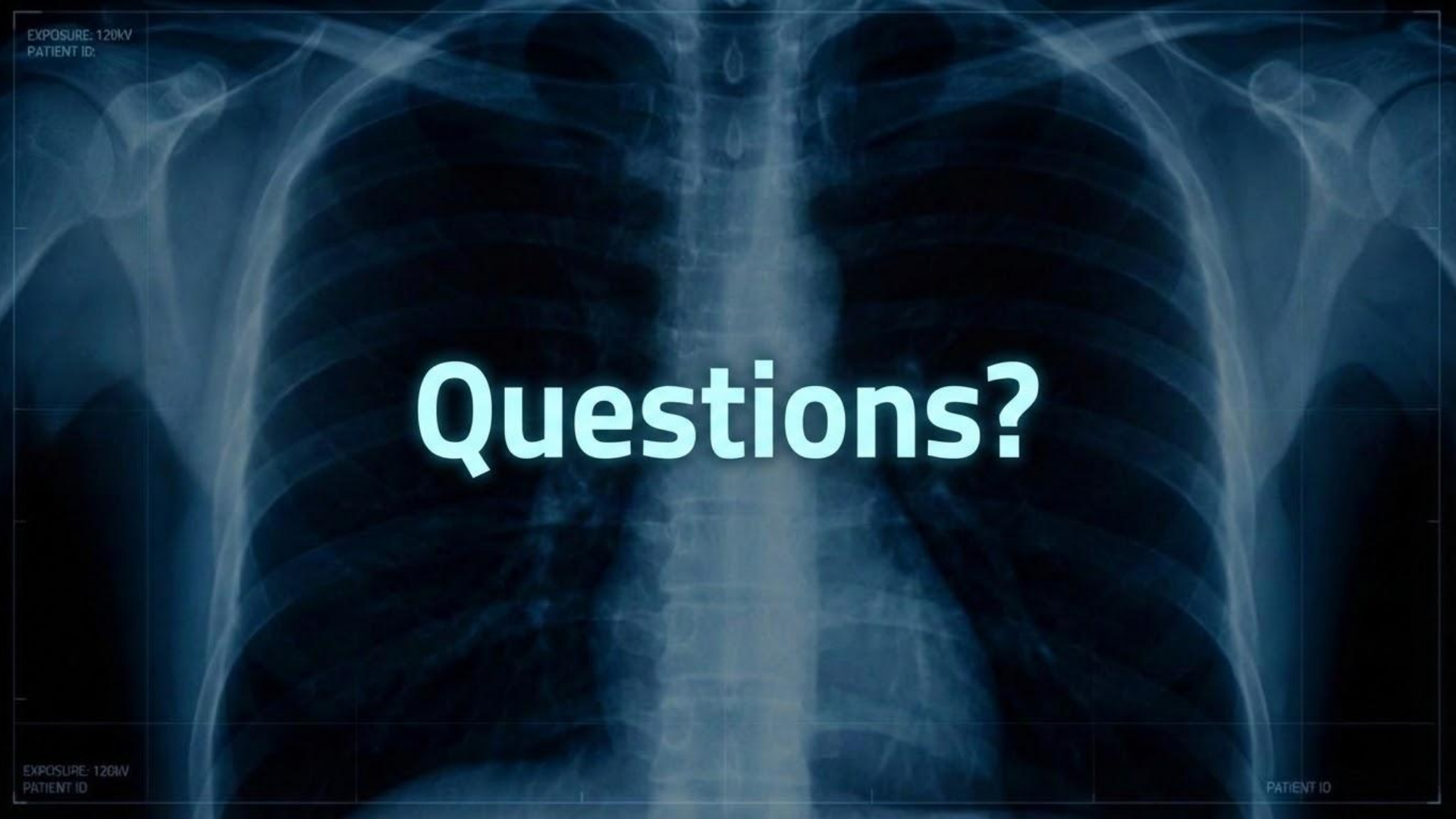
References

[1] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers, "ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017, pp. 3462–3471. doi: 10.1109/CVPR.2017.369.

[2] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA, Jul. 2017, pp. 4700–4708.

[3] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database,"in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Miami, FL, USA, Jun. 2009, pp. 248–255.

[4] GitHub Repository: https://github.com/pallavim8/Project-3-Cavalier-Kaggles.git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